

	Sanjivani Rural Educational Society's SANJIVANI COLLEGE OF ENGINEERING (An Autonomous Institution) Kopargaon – 423 603, Maharashtra.	ACAD-F-03
Academic Year: 2026-2027	Project Quality and Domain Categorization	Revision: 00 Dated :1/7/2026
Term – I	Department: Information Technology	Date of Preparation: 21/04/2026

Project Title: An Intelligent NLP System for Legal and Employment Document Risk Analysis with Multiplatform Delivery

Social Impact

High

Justification:

The project empowers users to understand complex legal and employment documents by identifying risky clauses, improving legal awareness, protecting user rights, and enabling informed decision-making.

Multidisciplinary Aspects

Yes

Justification:

The project integrates Artificial Intelligence, Natural Language Processing (NLP), Machine Learning, Legal Technology (LegalTech), Web Development, Database Management, and Cloud Computing.

Product-Based?

Yes

Justification:

The project delivers a deployable AI-powered web application and browser extension capable of analysing legal and employment documents in real time.

Quantum of Work

High

Justification:

The project involves literature review, dataset collection, NLP preprocessing, LLM integration, frontend and backend development, browser extension development, database integration, testing, deployment, and documentation.

Key Challenge(s)

- Accurate identification of legal clauses using NLP and LLMs.
- Processing multiple document formats (PDF, DOCX, URLs).
- Generating reliable risk and trust scores.
- Ensuring user data privacy and secure document handling.
- Optimizing response time for real-time document analysis.
- Minimizing AI hallucinations and improving recommendation accuracy

Remarks/Category

Innovative Project

Justification:

This project combines advanced AI, NLP, and LegalTech to build a deployable software solution with high social impact. It addresses a real-world problem through intelligent document analysis and provides practical assistance to users before accepting legal or employment agreements.

Project Team Roles and Responsibilities

1. Sushant More – AI/ML & NLP Engineer

Domain: Artificial Intelligence, Natural Language Processing (NLP)

Responsibilities:

- Literature survey on NLP and Large Language Models (LLMs)
- Data preprocessing and text cleaning
- Clause extraction and classification
- Risk analysis using LLMs (Llama 3/Mistral)
- Trust Score and Risk Score generation
- AI model evaluation and optimization

2. Varun Pawar – Backend & Database Developer

Domain: Backend Development and Database Management

Responsibilities:

- Develop REST APIs using FastAPI
- Document upload and text extraction services
- MongoDB/Firebase database design and integration
- User authentication and authorization
- Report generation and history management
- API security and backend deployment

3. Pratiksha Rahane – Frontend & Browser Extension Developer

Domain: Frontend Development

Responsibilities:

- Design and develop the React.js web application
- Create responsive UI using Tailwind CSS
- Develop Chrome Browser Extension
- Dashboard development and visualization
- Display AI analysis, summaries, and recommendations
- Integrate frontend with backend APIs

4. Payal Shinde – Project Coordinator, Research & Testing Engineer

Domain: Software Engineering, Quality Assurance, Project Management

Responsibilities:

- Requirement analysis and SRS preparation
- System design, UML diagrams, and documentation
- Project planning and progress monitoring
- Functional, integration, and performance testing
- Project report, research paper, PPT, and final documentation
- Cloud deployment support and project presentation

Mr. D. R. Cholke
Name and Signature of Project Guide

Mr. Sushant More
Mr. Varun Pawar
Miss. Pratiksha Rahane
Miss. Payal Shinde
Name and Signature of Group Members

